

LAB 6

Solve below Questions:

1. Write the programme to sort the integers 8, 4, 3,5,6 and the alphabetical string C, O, I, P, U, in ascending order. Show the resulting output.

PROGRAM:

```
package Lab6;

import java.util.Arrays;

public class SortedElements {

    public static void main(String[] args) {

        // Sorting integers
        int[] intArr = {8, 4, 3, 5, 6};
        Arrays.sort(intArr);
        System.out.println("Sorted integers: " + Arrays.toString(intArr));

        // Sorting strings
        String[] strArr = {"C", "O", "I", "P", "U"};
        Arrays.sort(strArr);
        System.out.println("Sorted strings: " + Arrays.toString(strArr));

    }
}
```

OUTPUT:

```
<terminated> SortedElements [Java Application] C:\Users\A
Sorted integers: [3, 4, 5, 6, 8]
Sorted strings: [C, I, O, P, U]
```

2. Write a Java program to implement the bubble sort algorithm to sort an array of integers in ascending order.

PROGRAM:

```
package mDemo;

public class BubbleSortArray {

    public static void main(String[] args) {
        // Array of integers
        int nums[] = {5, 6, 12, 45, 7, 87, 22, 1, 0 };
        int size = nums.length; // Size variable to store the size of the
array
        int temp = 0; // Temporary Variable for Swapping of
variable
```

```

C:\Users\Ashi\OneDrive\p2\p001\
Sorting stages:
5 6 12 7 45 22 1 0 87
5 6 7 12 22 1 0 45 87
5 6 7 12 1 0 22 45 87
5 6 7 1 0 12 22 45 87
5 6 1 0 7 12 22 45 87
5 1 0 6 7 12 22 45 87
1 0 5 6 7 12 22 45 87
0 1 5 6 7 12 22 45 87
0 1 5 6 7 12 22 45 87
After Bubble Sort:
0 1 5 6 7 12 22 45 87
The largest element of array is : 87

```

3. Write a program to input an array 10 elements and print the cube of prime numbers in it.

PROGRAM:

```
package Lab6;

import java.util.Scanner;

public class PrimeNumberCubeArray {

    public static void main(String[] args) {
        // Creating a Scanner object for user input
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the size of array: ");
        int n = sc.nextInt();

        // Initialize the array with size n
        int[] nums = new int[n];

        // Read elements into the array
        for (int i = 0; i < n; i++) {
            System.out.print("Enter element " + (i + 1) + ": ");
            nums[i] = sc.nextInt();
        }

        System.out.println("Prime numbers in the array:");

        // Check for prime numbers and print them
        for (int i = 0; i < n; i++) {
            boolean isPrime = true;

            // Check if the number is prime
            if (nums[i] <= 1) {
                isPrime = false;
            } else {
                for (int j = 2; j <= Math.sqrt(nums[i]); j++) {
                    if (nums[i] % j == 0) {
                        isPrime = false;
                        break;
                    }
                }
            }

            // Print the number if it is prime
            if (isPrime) {
                System.out.println(nums[i]);
            }
        }

        sc.close();
    }
}
```

OUTPUT:

```
<terminated> PrimeNumberCubeArray [Java Application] C:\Users\Ashi
Enter the size of array: 5
Enter element 1: 1
Enter element 2: 2
Enter element 3: 3
Enter element 4: 4
Enter element 5: 5
Prime numbers in the array:
2
3
5
```

4. Write a java program to implement integer wrapper class methods.(any 3 methods)

PROGRAM:

```
package Lab6;

public class WrapperClassMethods {
    public static void main(String[] args) {

        int a = 100;
        // Method 1 - Integer.valueOf()
        Integer num1 = Integer.valueOf(a);          // Integer.valueOf()
        assigns value to a wrapper class object
        System.out.println("Integer num1: " + num1);

        // Method 2 - intValue()
        int intPrimitive = num1.intValue();          // This function converts
        wrapper class object value to int value
        System.out.println("Primitive int value obtained from Integer
        object: " + intPrimitive);

        // Method 3 - parseInt(String)
        String numStr = "73739338";
        int parsedInt = Integer.parseInt(numStr);    // Converts Number
        String value into integer
        System.out.println("Parsed integer from string \"" + numStr + "\": "
        + parsedInt);
    }
}
```

OUTPUT:

```
<terminated> WrapperClassMethods [Java Application] C:\Users\Ashi chouhan\p2\pc
Integer num1: 100
Primitive int value obtained from Integer object: 100
Parsed integer from string "73739338": 73739338
```

5. Write a java program to implement double wrapper class methods.(any 3 methods)

PROGRAM:

```
package Lab6;

public class DoubleWrapperClassMethods {

    public static void main(String[] args) {

        // Method 1 - valueOf(double d)
        Double doubleNum = Double.valueOf(25.78);    // Assigns value to a
double wrapper class object
        System.out.println("Double object created with valueOf: " +
doubleNum);

        // Method 2 - doubleValue()
        double doublePrimitive = doubleNum.doubleValue();    // Wrapper
class object value into primitive value
        System.out.println("Primitive double value obtained from Double
object: " + doublePrimitive);

        // Method 3 - parseDouble(String s)
        String doubleString = "27483.264";
        double parsedDouble = Double.parseDouble(doubleString);    //
Converts Double stirng value into a double value
        System.out.println("Parsed double from string \" + doubleString +
\"\": \" + parsedDouble);
    }
}
```

OUTPUT:

```
Double object created with valueOf: 25.78
Primitive double value obtained from Double object: 25.78
Parsed double from string "27483.264": 27483.264
```

6. Write a java program to implement float wrapper class methods.(any 3 methods)

PROGRAM:

```
package Lab6;

public class FloatWrapperClassMethods {

    public static void main(String[] args) {

        // Method 1 - valueOf(float f)
        Float floatNum = Float.valueOf(10.75f); // Assigns value to a float
wrapper class object
        System.out.println("Float object created with valueOf: " + floatNum);

        // Method 2 - floatValue()
```

```

        float floatPrimitive = floatNum.floatValue(); // Wrapper class object
        value into primitive value
        System.out.println("Primitive float value obtained from Float object: " +
floatPrimitive);

        // Method 3 - parseFloat(String s)
        String floatString = "123.456";
        float parsedFloat = Float.parseFloat(floatString); // Converts float
        string value into a float value
        System.out.println("Parsed float from string \"" + floatString + "\": " +
parsedFloat);
    }
}

```

OUTPUT:

```

<terminated> FloatWrapperClassMethods [Java Application] C:\Users\Ashi chouhan\.p2\pool\plu
Float object created with valueOf: 10.75
Primitive float value obtained from Float object: 10.75
Parsed float from string "123.456": 123.456

```

7. Write a Java program to validate email addresses using regular expressions. The email should have the format username@domain.com where username and domain can contain alphanumeric characters, dots, and hyphens.

PROGRAM:

```

package Lab6;

import java.util.Scanner;
import java.util.regex.Pattern;

public class RegexValidateEmail {

    public static void main(String[] args) {
        // Regex for email
        String emailRegex = "^[a-zA-Z0-9._-]+@[a-zA-Z0-9.-]+\\.([a-zA-Z]{2,})$";

        // Compiling the regex
        Pattern p = Pattern.compile(emailRegex);

        Scanner sc = new Scanner(System.in);

        // Taking user input for email using scanner object
        System.out.print("Enter an email address: ");
        String email = sc.nextLine();

        // Boolean variable to check if entered email matches the regex pattern
        boolean isValidEmail = p.matcher(email).matches();

        // Printing valid and invalid email based on boolean variable value
        if (isValidEmail) {
            System.out.println(email + " is a valid email address.");
        }
    }
}

```

```

    } else {
        System.out.println(email + " is not a valid email address.");
    }

    sc.close();
}
}

```

OUTPUT:

```

<terminated> RegexValidateEmail [Java Application] C:\Users\Ashi ct
Enter an email address: ashi123@abc.com
ashi123@abc.com is a valid email address.

```

8. Create a Java program to validate phone numbers. The format should be (xxx) xxx-xxxx where x is a digit.

PROGRAM:

```

package Lab6;

import java.util.Scanner;
import java.util.regex.Pattern;

public class RegexValidatePhoneNumber {

    public static void main(String[] args) {
        // Regex for phone number
        String phoneRegex = "^\\(\\d{3}\\) \\d{3}-\\d{4}$";

        // Compiling the regex
        Pattern p = Pattern.compile(phoneRegex);

        Scanner sc = new Scanner(System.in);

        // taking user input for phone number using scanner object
        System.out.print("Enter a phone number in the format (xxx) xxx-xxxx: ");
        String phoneNumber = sc.nextLine();

        // Boolean variable to check if phone number matches the regex pattern
        boolean isValid = p.matcher(phoneNumber).matches();

        // Printing valid or invlaid based on boolean variable pattern
        if (isValid) {
            System.out.println(phoneNumber + " is a valid phone number.");
        } else {
            System.out.println(phoneNumber + " is not a valid phone number.");
        }

        sc.close();
    }
}

```

OUTPUT:

```
<terminated> regexvalidatephonenumber.java Application] C:\Users\Ashir Choudhary\AppData\Local\Programs\Java\jdk-11.0.2\bin\java.exe
Enter a phone number in the format (xxx) xxx-xxxx: (123) 456-7890
(123) 456-7890 is a valid phone number.
```